

AUTOMOBILE INSURANCE COMPLAINTS RANKINGS

INTERNSHIP PROJECT REPORT

Project Code: CDACL-001

Submitted By

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DECLARATION

I hereby declare that the project report entitled "Automobile Insurance Complaints Rankings and Premium Trend Analysis" is an original work carried out by me during the internship period. The analysis, visualizations, findings, and conclusions presented in this report are based on the provided dataset and my own efforts.

I further declare that this project has not been submitted, either in part or in full, to any other institution or organization for the award of any degree, diploma, or certification. All sources of information used in this report have been acknowledged appropriately.

This project has been completed in accordance with the guidelines provided by the internship organization and reflects my understanding and application of data analytics concepts, SQL queries, and Power BI dashboard development.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my internship mentor, project guide, and organization for providing me with the opportunity to work on this project. Their continuous support, valuable guidance, and constructive feedback played a significant role in the successful completion of this study.

I am especially thankful for the knowledge and practical exposure gained throughout the internship. The guidance provided during each stage of the project helped me strengthen my understanding of data analytics, business intelligence, SQL, and Power BI.

I would also like to thank my colleagues and team members for their cooperation, encouragement, and willingness to share their knowledge whenever required. Their support contributed greatly to overcoming challenges encountered during the project.

Finally, I extend my heartfelt appreciation to everyone who contributed directly or indirectly to the successful completion of this project. This internship has been a valuable learning experience that enhanced both my technical expertise and professional skills, preparing me for future opportunities in the field of data analytics.

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ABSTRACT

This project analyses automobile insurance complaints and premium trends of insurance companies in New York State. The main objective is to understand customer complaint patterns, evaluate complaint ratios, analyse premium growth, and measure company performance.

Power BI dashboards were used to create interactive visualizations for complaint trends, company rankings, premium analysis, and key performance indicators. The analysis helped identify complaint patterns, peak complaint periods, top-performing companies, and companies with the highest premium values.

The results show that better claims handling, faster customer support, and lower complaint ratios lead to higher customer satisfaction and improved business performance. This project highlights how data analytics and business intelligence tools can support decision-making and help insurance companies improve service quality and operational efficiency.

1. INTRODUCTION

Automobile insurance helps vehicle owners protect themselves from financial loss caused by accidents, theft, vehicle damage, and other unexpected situations. Today, as competition in the insurance industry continues to grow, providing good customer service and maintaining customer satisfaction have become more important than ever.

Customer complaints are one of the best ways to understand how well an insurance company performs. The way a company handles claims, resolves issues, and supports customers has a direct impact on its reputation and customer trust. Companies with fewer complaints usually provide better service and manage customer concerns more effectively.

This project focuses on analysing automobile insurance complaint rankings and premium trends in New York State using data analytics and Power BI dashboards. The analysis includes data from 223 insurance companies, around 36,000 customer complaints, and total premiums of 87.43K million dollars.

The main purpose of this study is to identify complaint trends, evaluate company performance, understand premium growth, and generate useful insights that can help improve customer satisfaction and overall business performance.

2. OBJECTIVES

The main goal of this project is to analyse automobile insurance complaint data and premium trends to gain insights into customer satisfaction and the performance of insurance companies. The project analyses complaint records, premium growth, and company rankings to identify critical trends and produce valuable business insights.

The purpose of this study is to investigate the impact of various complaint types on insurance companies and the effect of complaint ratios on their ranking. It also looks at premium growth over the years to pinpoint trends and factors that contribute to changes in premium values.

3. DATASET DESCRIPTION

The dataset contains automobile insurance complaint information and premium statistics for insurance companies operating in New York State.

Dataset Attributes

Attribute	Description
NAIC	Insurance Company ID
Company Name	Name of Insurance Company
Ratio	Complaint Ratio
Upheld Complaints	Valid Complaints
Question of Fact Complaints	Investigation Cases
Not Upheld Complaints	Rejected Complaints
Total Complaints	Total Complaint Count
Premiums Written	Premium Amount
Rank	Company Rank
Filing Year	Year of Complaint Closure

Dataset Summary

- Total Companies: 223
- Total Premiums: 87.43K
- Total Complaints: 36K
- Average Complaint Ratio: 1.89

4. METHODOLOGY

The project followed a structured data analytics approach.

Data Collection

The dataset was obtained from the provided insurance database.

Data Cleaning

- Removed duplicate records.
- Handled missing values.
- Standardized company names and data formats.

Data Analysis

- Complaint analysis.
- Premium trend analysis.
- Company performance comparison.

Dashboard Development

Interactive dashboards were created using Power BI to visualize business insights.

Power BI Dashboard Design

Dashboard Objectives

The dashboard was created to give a clear and complete view of insurance company performance, customer complaint trends, and premium growth over time.

Dashboard Features

Interactive Filters

- Company Name Filter
- Filing Year Filter

KPI Cards

- Total Companies
- Total Complaints
- Total Premiums
- Average Complaint Ratio

Interactive Visualizations

- Pie Charts
- Bar Charts
- Line Charts
- Funnel Charts

5. COMPLAINT ANALYSIS

The complaint analysis categorized complaints into:

Not Upheld Complaints

Approximately 20,229 complaints were classified as not upheld, making this the largest complaint category.

Question of Fact Complaints

Approximately 11,474 complaints required additional investigation and verification.

Upheld Complaints

Approximately 4,216 complaints were upheld and confirmed as valid customer grievances.

Complaint Trend Analysis

The year 2009 recorded the highest complaint volume and was identified as the peak complaint year. Complaint levels gradually decreased in later years, indicating improvements in claims processing and customer service practices.

Best and Worst Ranked Companies

- Best Ranked Company: Progressive Specialty Insurance
- Worst Ranked Company: Zurich American Insurance Company

These rankings were determined based on complaint ratios and complaint management performance.

6. PREMIUM TREND ANALYSIS

Premium analysis was conducted to evaluate business growth and pricing trends.

Key Metrics

- Total Premiums: 87.43K
- Premium Growth: 612.13%
- Top Premium Company: GEICO General Insurance Company

Premium Trend Findings

The analysis revealed a consistent increase in premium values over the reporting period. Premium growth reached 612.13%, indicating substantial expansion in insurance business operations.

7. COMPANY COMPARISON

Five major insurance companies were analysed:

1. GEICO General Insurance Company
2. Government Insurance Company
3. Allstate Insurance Company
4. Victoria Automobile Insurance Company
5. American Modern Home Insurance Company

Comparison Metrics

- Complaint Ratio
- Premium Volume
- Total Complaints
- Company Ranking

Analysis

GEICO General Insurance Company emerged as the leading company in premium volume. Companies with lower complaint ratios consistently achieved better rankings and customer satisfaction levels.

8. DASHBOARD OVERVIEW

The Power BI dashboard consists of four analytical pages:

Executive Summary

- Total Companies
- Total Complaints
- Total Premiums
- Average Complaint Ratio

Complaint Analysis

- Complaint Distribution
- Complaint Trends
- Best and Worst Ranked Companies

Premium Analysis

- Premium Growth Trends
- Premium Share Analysis
- Top Premium Companies

Company Comparison

- Complaint Ratio Comparison
- Premium Comparison

9. KEY FINDINGS

1. The dataset contains 223 automobile insurance companies.
2. Total complaints recorded were approximately 36,000.
3. Total premiums reached 87.43K.
4. The average complaint ratio was 1.89.
5. The year 2009 recorded the highest complaint volume.
6. Approximately 4,000 complaints were upheld.
7. GEICO General Insurance Company generated the highest premium volume.
8. Great Northern Insurance Company achieved the best ranking.
9. Premium growth reached 612.13% during the analysis period.
10. Lower complaint ratios were strongly associated with higher company rankings.

10. RECOMMENDATIONS

Based on the analysis results, the following actions are recommended to improve insurance company performance and customer experience:

1. Speed Up Claim Processing

Reducing claim settlement time can improve customer satisfaction and lower complaint rates.

2. Provide Better Customer Support

Quick responses and clear communication can help customers feel informed and supported.

3. Review Complaints Frequently

Regularly analysing customer complaints helps identify problems early and improve service quality.

4. Expand Digital Services

Online claim submission, mobile applications, and digital support can make services easier and faster for customers.

5. Use Data for Business Improvement

Data analysis can help companies understand trends, predict future issues, and make better business decisions.

6. Maintain Clear Policies

Simple and transparent policy information can reduce confusion and increase customer trust.

CONCLUSION

The Automobile Insurance Complaints Rankings and Premium Trend Analysis project successfully analysed customer complaints, premium growth, and overall company performance in New York State's automobile insurance industry.

The analysis showed that complaint ratios have a strong impact on company rankings and customer satisfaction. Insurance companies with faster claims processing and better customer service generally achieved stronger performance and lower complaint rates.

The study also found that GEICO General Insurance Company recorded the highest premium volume, while Erie Insurance Company of New York achieved the best ranking based on complaint performance.

Overall, this project demonstrates how data analytics and Power BI can help insurance companies understand customer behaviour, improve decision-making, and enhance service quality for long-term business growth.

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